Suggested Bibliography

By Spyros Tserkis

January 31, 2026

Quantum Mechanics

- McIntyre - “Quantum Mechanics: A Paradigms Approach”
- Sakurai, Napolitano - “Modern Quantum Mechanics”
- Ballentine - “Quantum Mechanics: A Modern Development”
- Peres - “Quantum Theory: Concepts and Methods”
- Bohm - “Quantum Mechanics: Foundations and Applications”

Quantum Optics

- Grynberg, Aspect, Fabre - “Introduction to Quantum Optics”
- Walls, Milburn - “Quantum Optics”

Mathematical Structure of Quantum Theory

- Heinosaari, Ziman - “The Mathematical Language of Quantum Theory”
- Prugovecki - “Quantum Mechanics in Hilbert Space”
- Hall - “Quantum Theory for Mathematicians”
- Strocchi - “An Introduction to the Mathematical Structure of Quantum Mechanics”
- Moretti - “Spectral Theory and Quantum Mechanics”

History & Interpretation of Quantum Mechanics

- Jammer - “The Philosophy of Quantum Mechanics”
- Plotnitsky - “Reading Bohr: Physics and Philosophy”

Quantum Information Theory

- Ekert, Hosgood, Kay, Macchiavello - “Introduction to Quantum Information Science”
- Watrous - “Understanding Quantum Information and Computation”
- Nakahara, Ohmi - “Quantum Computing”
- Nielsen, Chuang - “Quantum Computation and Quantum Information”
- Riccardo, Motta - “Quantum Information Science”
- Holevo - “Quantum Systems, Channels, Information”

Classical Information Theory

- Franceschetti - “Wave Theory of Information”
- Cover, Thomas - “Elements of Information Theory”

Probability Theory & Statistics

- Bertsekas, Tsitsiklis - “Introduction to Probability”
- Rényi - “Probability Theory”
- Rényi - “Foundations of Probability”

Matrix Analysis

- Mayer - “Meyer Matrix Analysis And Applied Linear Algebra”
- Garcia, Horn - “Matrix Mathematics: A Second Course in Linear Algebra”
- Horn, Johnson - “Matrix Analysis”

Linear Algebra

- Axler - "Linear Algebra Done Right"
- Jänich - "Linear Algebra"
- Halmos - "Finite-Dimensional Vector Spaces"
- Hoffman, Kunze - "Linear Algebra"
- Roman - "Advanced Linear Algebra"

Mathematical Analysis

- Apostol - "Mathematical Analysis"
- Rudin - "Principles of Mathematical Analysis"

Real Analysis

- Kolmogorov, Fomin, Silverman - "Introductory Real Analysis"
- Axler - "Measure, Integration & Real Analysis"

Functional Analysis

- Halmos - "Introduction to Hilbert Space and the Theory of Spectral Multiplicity"
- Reed, Simon - "Functional Analysis. Vol I: Functional Analysis"
- Rudin - "Functional Analysis"

Operator Algebras

- Bratteli, Robinson - "Operator Algebras and Quantum Statistical Mechanics"
- Blackadar - "Operator Algebras"

Differential Equations

- Arnold - "Ordinary Differential Equations"
- Arnold - "Lectures on Partial Differential Equations"

Classical Mechanics

- Thorton, Marion - "Classical Dynamics"
- Arnold - "Mathematical Methods of Classical Mechanics"

Chaotic Dynamical Systems

- Devaney - "A First Course In Chaotic Dynamical Systems"
- Devaney - "An Introduction To Chaotic Dynamical Systems"
- Hirsch, Smale - "Differential Equations, Dynamical Systems, and Linear Algebra"
- Kowalski, Steeb - "Nonlinear Dynamical Systems and Carleman Linearization"

Logic & Computability

- Jeffey - "Formal Logic: Its Scope and Limits"
- Enderton - "A Mathematical Introduction to Logic"
- Cutland - "Computability: An Introduction to Recursive Function Theory"
- Boolos, Burgess, Jeffrey - "Computability and Logic"